

Digital Logic Circuits

Week1: Introduction

Professor: Jasurbek Khodjaev
IUT, Spring 2024
SOC2080





Info

- Professor: Jasurbek Khodjaev
- Office: B412 (Left wing)
- email: j.khodjaev@inha.uz
- T.A. : TBA
- Office hours: TBA



Course info

- Text:
 - Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design (3rd Ed.), McGraw-Hill, 2014.
 - Ref:
 - Alan B. Marcovitz, Introduction to Logic Design (3rd Ed.), McGraw-Hill, 2010.
 - Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design (2nd Ed.), McGraw-Hill, 2008.
- Grading : *subject to change*
 - Attendance **10%**, Homeworks **10%**, Term Project **10%**
 - Midterm Exam **35%**, Final Exam **35%**
- Any academic misconduct will result in zero.
- Attendance after 5~10 minutes!!!



Telegram channel

t.ly/hS1MY



Course info

- Homework Policies
 - All homework are to be done by each student individually
 - **Late homework will not be accepted (*whatever the reason, including medical*)**
 - Homework should be submitted into the e-class





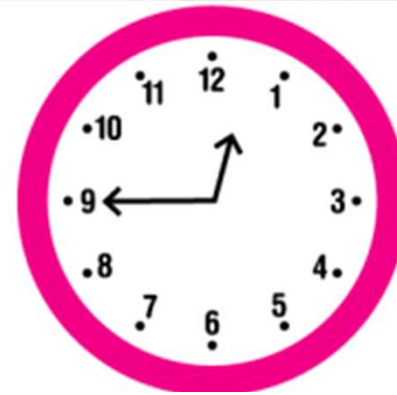
Analog vs. Digital

- Data/Information can be **analog** or **digital**
- The term **analog data** refers to information that is continuous
 - Analog data take on continuous values
- **Digital data** refers to information that has discrete states
 - Digital data take on discrete values.

Analog vs. Digital



12:45:12



12:45:12:125548...

Analog Signal



Digital Signal



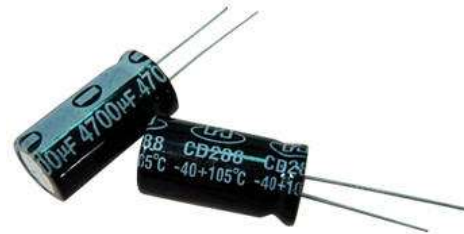
Analog vs. Digital

- Analog circuit composed of the basic elements as:

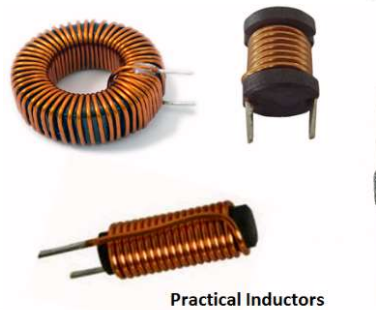
- Resistor



- Capacitor



- Inductor



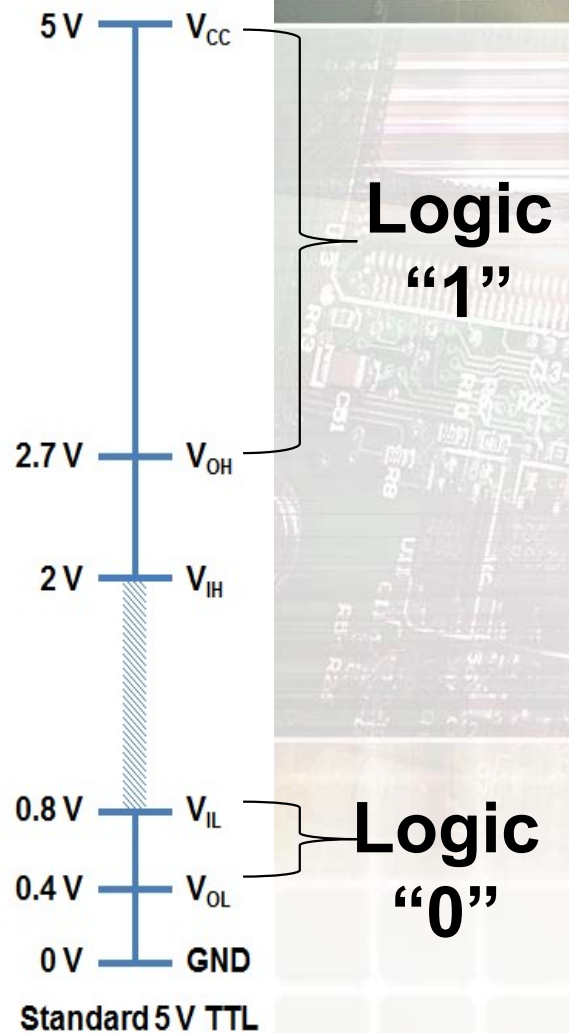
Practical Inductors

- Etc.

Analog vs. Digital



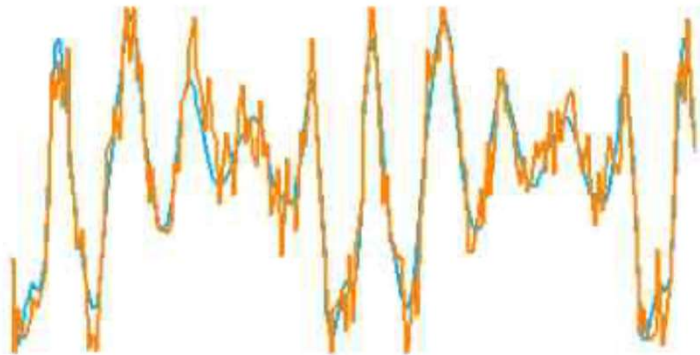
Analog vs. Digital



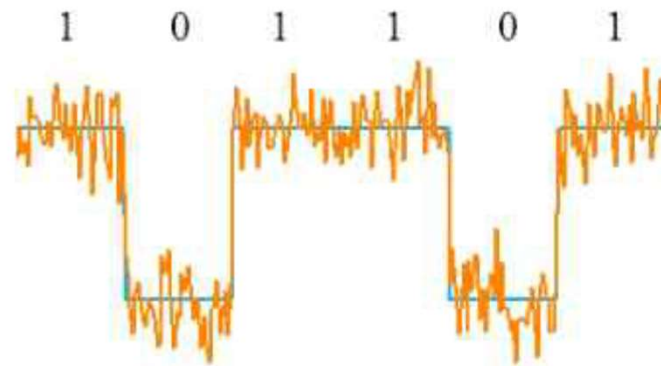
- A digital circuit is composed of basic elements as:
 - AND, OR, NAND and NOR logic gates
- Recognizes and produces only two voltage level
 - High (+5v) or Low (0V)

Why Digital?

- One of the most important advantage of digital system is noise immunity
- Receiver circuitry can distinguish between a binary 0 and 1 with a significant amount of noise

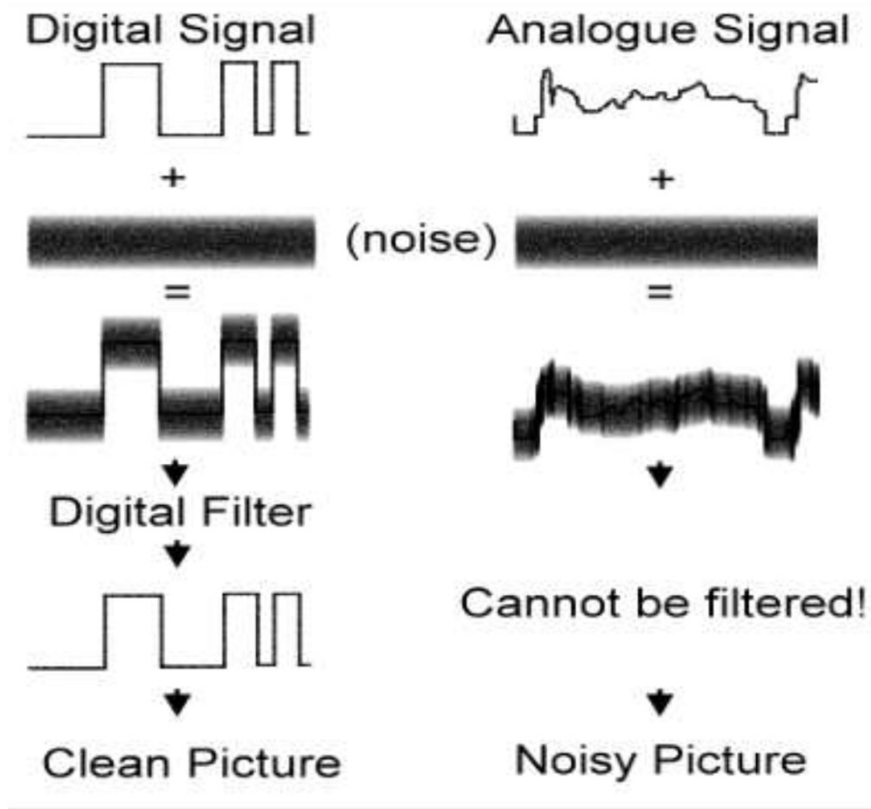


analog signal with noise



digital signal with noise

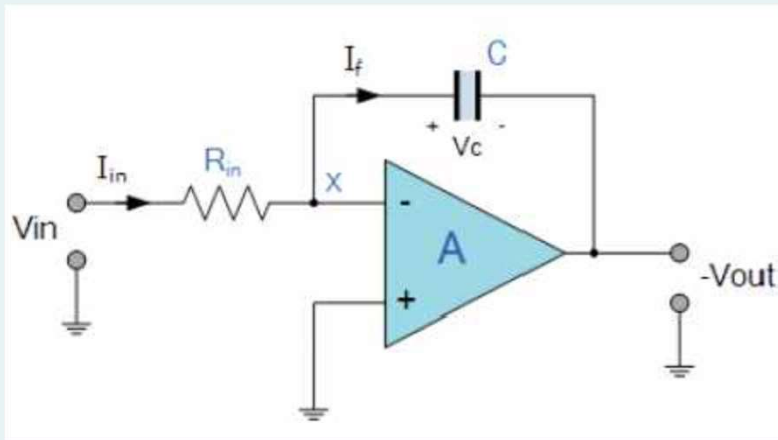
Why Digital?



Why Digital?

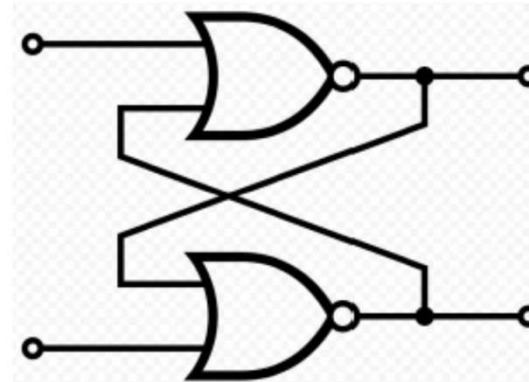
- Memory

Memory for analog computer



Leakage (NO ideal capacitor), Big size

Memory for digital computer



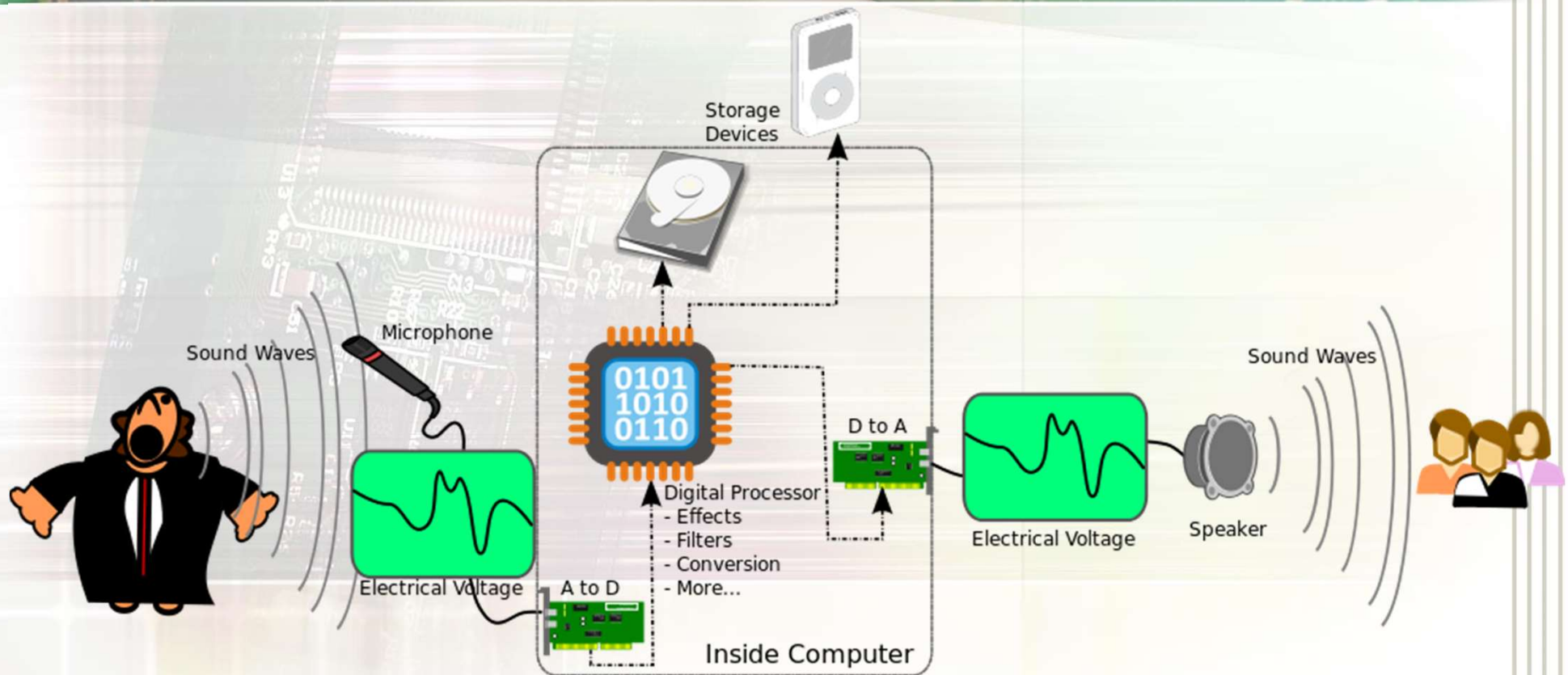
No leakage, Small size



Pros and cons of Digital system

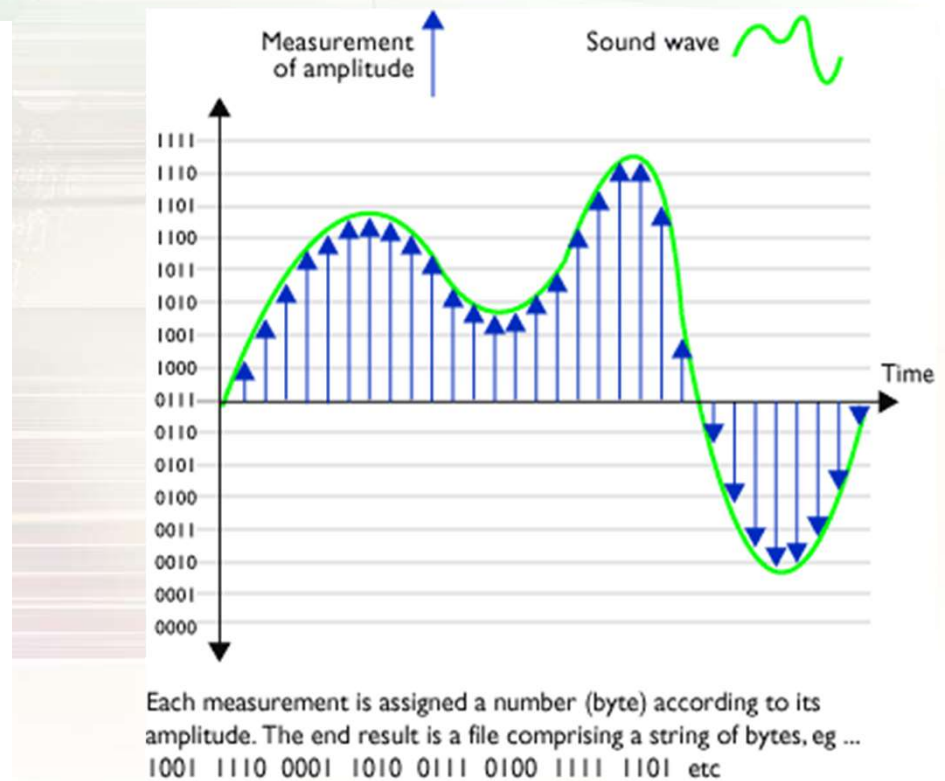
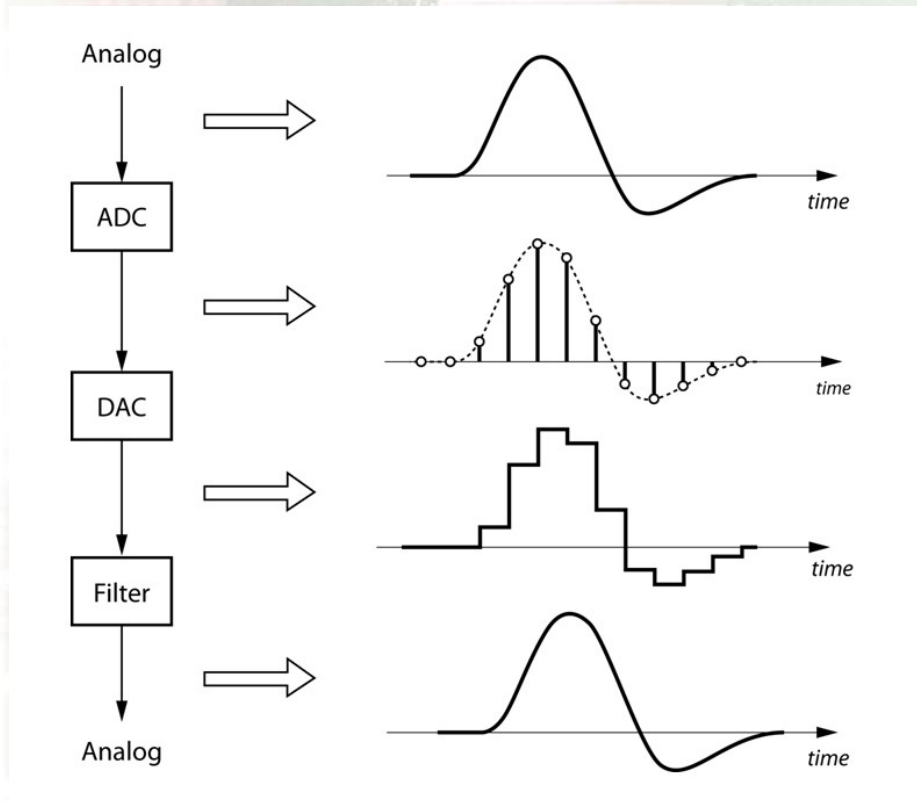
- **Reproducibility**
 - Given the same inputs, digital circuit generates the same outputs
- **Ease of design**
 - No complicated math skills are needed
 - The behavior of small circuits can be understood without knowing the details of complicated devices
- **Flexibility and functionality**
 - Different way to process digitalized data (compress, encrypt, store)
- **Programmability**
 - Hardware description language to design circuit
- **Speed**
 - Very fast speed. Several gigahertz clock rate
 - (✕ the speed of analog circuits ? light speed)
- **Economy/compactness**
 - Very small. Millions of transistors on a chip
- **Rapidly and steadily advancing technology**
- **Cons:**
 - The real world is analog
 - Converting data to digitized signals takes time

Analog ↔ Digital



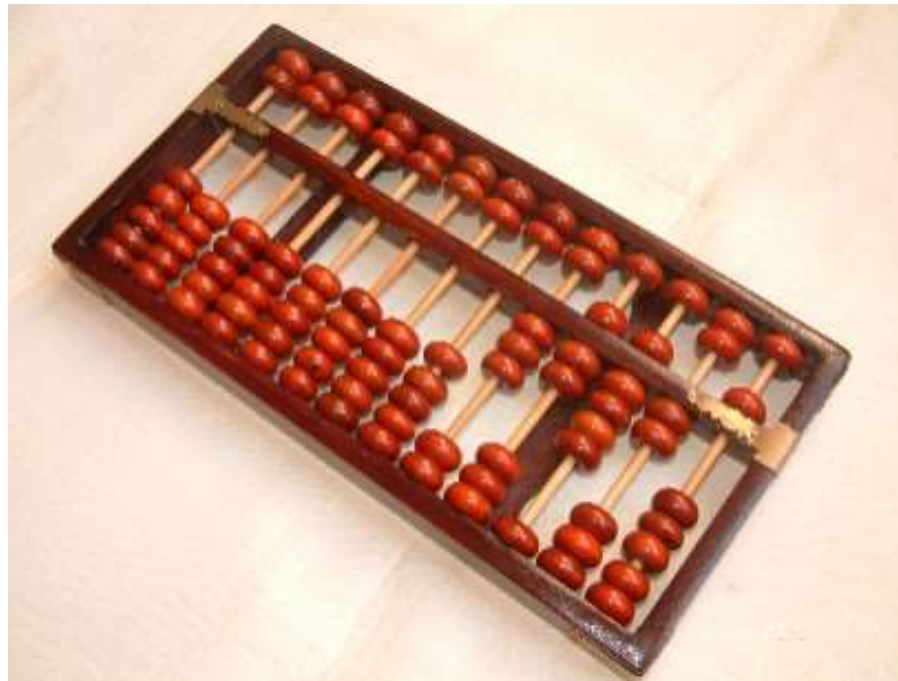
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Analog ↔ Digital



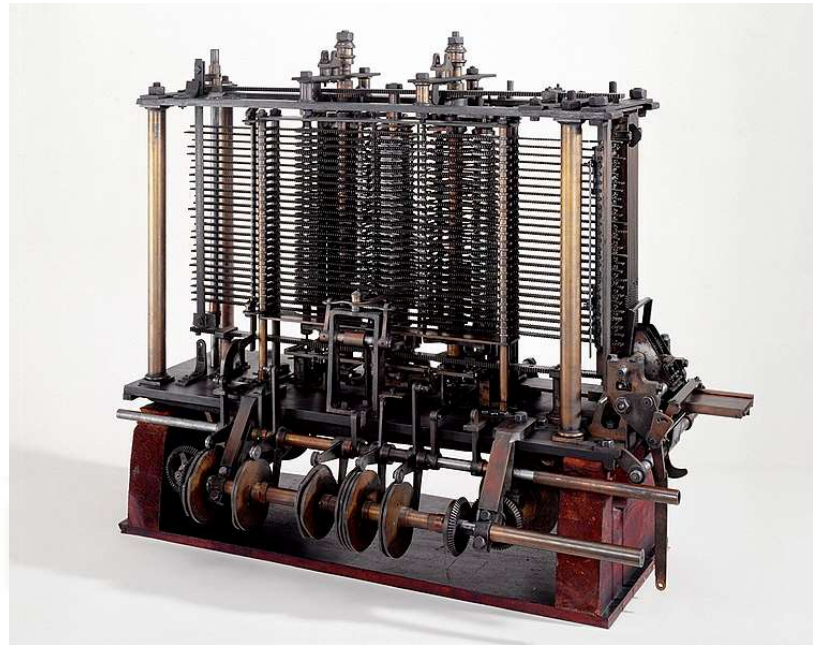
Analog / Digital computer

- Abacus (China, 500 B.C.)



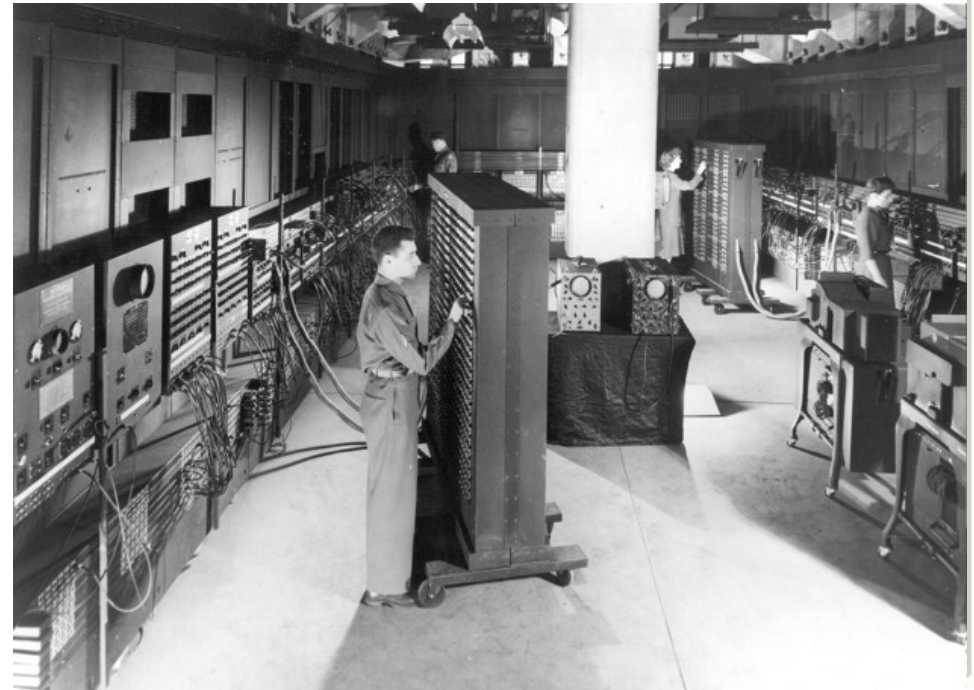
Analog / Digital computer

- The first mechanical computer
- Charles Babbage machine
 - Difference Engine 1 (1837)
 - Consists of 25000 mechanical parts
 - Two-cycle sequence: store & execute



Analog / Digital computer

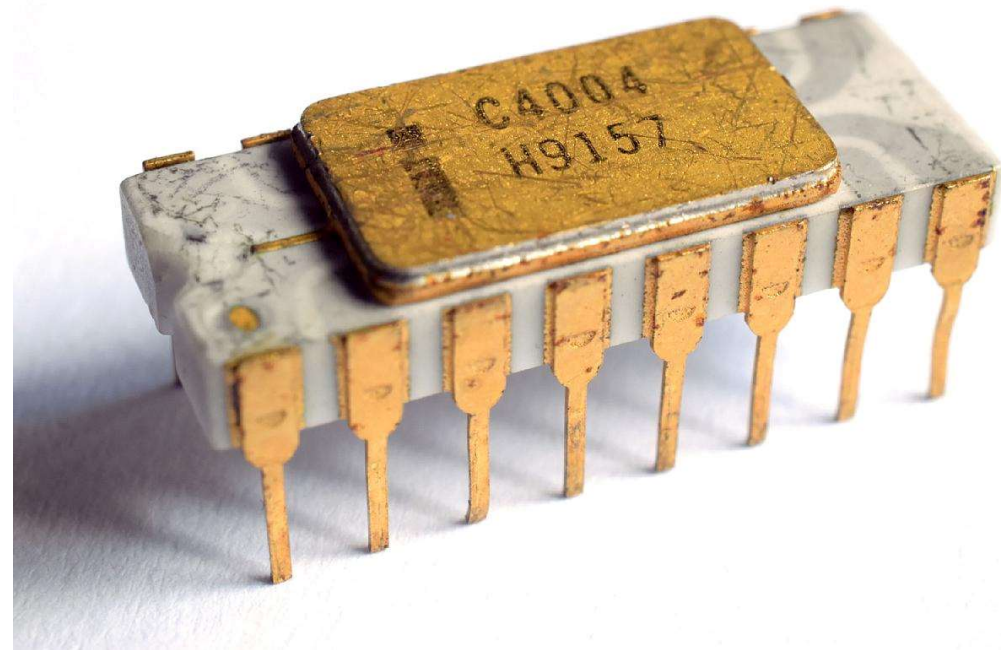
- First electronic digital computer
 - Electronic Numerical Integrator And Computer (ENIAC)
 - University of Pennsylvania, 1945
 - Size:
 - 24m long x 2.6m high
 - Used 18000 vacuum tubes



Processor chips

- **Intel 4004 - 1971**

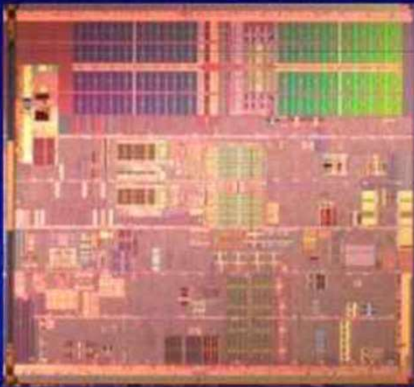
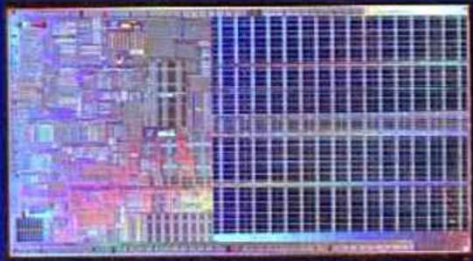
- The first commercially available microprocessor by Intel
- 2'250 transistors
- CPU: 740 kHz
- 10 μm



Processor chips

- Intel introduced 90nm technology based chips, 2003

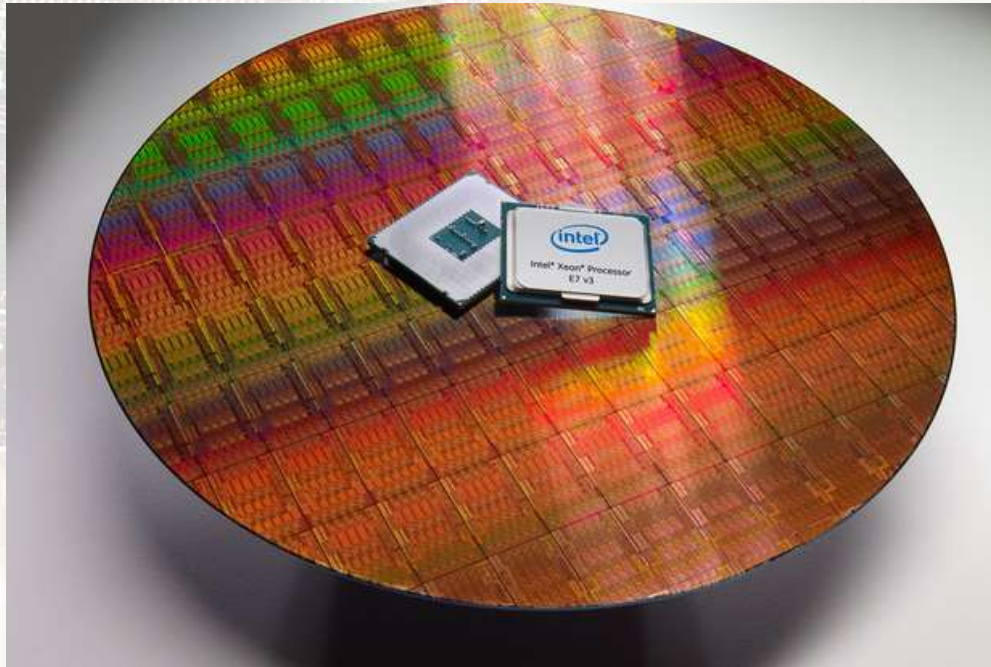
90 nm CPU Chips

Prescott CPU	Dothan CPU
	
112 mm ² die size 125 million transistors	87 mm ² die size 144 million transistors

90 nm process now ramping on high performance CPU products

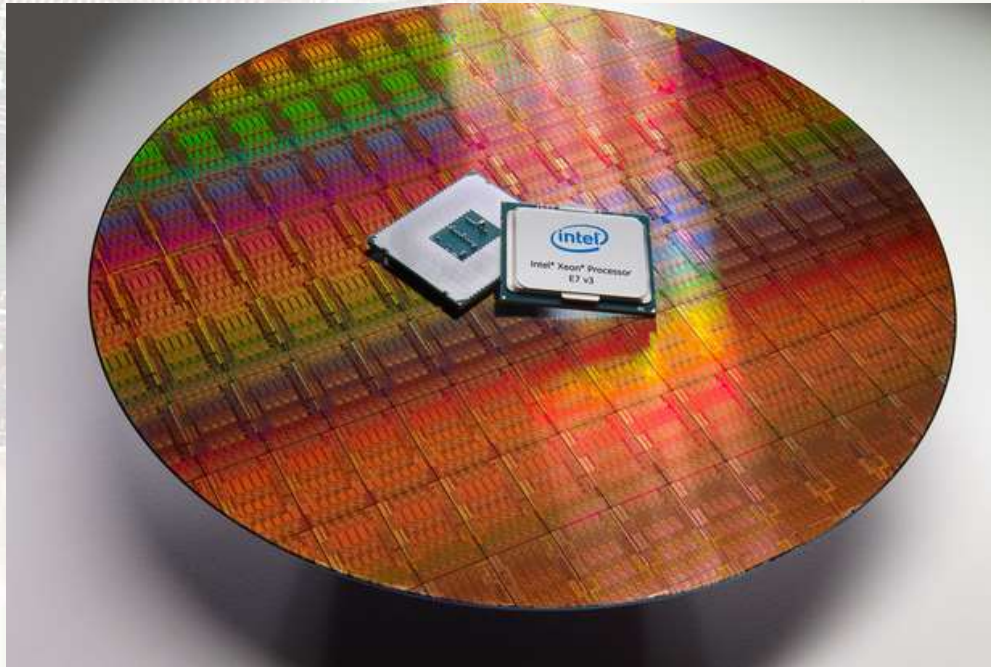
Processor chips

- Intel “Skylake” 14nm technology based chip, 2015
 - 1.75 bn transistors



Processor chips

- Intel “Alder Lake” 10nm technology based chip, 2022
 - 4.3 bn transistors

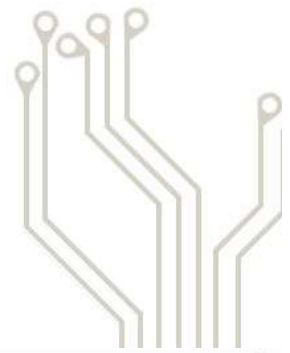




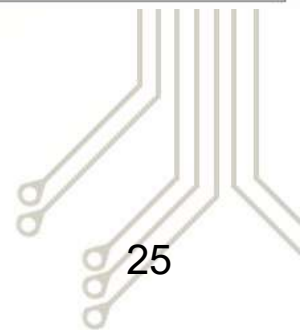
Processor chips

- In May 2021, IBM announced it had produced **2 nm chipmaking technology** and had successfully made a prototype "fingernail-sized" chip with upwards of **50 billion transistors**, [wikipedia]
- which translates to **333 million transistors per square millimeter**
- But you know what is really amazing - how those billion transistors work without defects!

1958 vs 2015



Measure	Elliott 405	Raspberry Pi Zero
		
Year	1957	2015
Price	£85,000 (1957)	\$5
Instruction cycle time	10.71-0.918 ms (93-1089 Hz)	1 ns (1 GHz clock)
Main memory	16 kB drum store	512 MB LPDDR2 SDRAM
Fast memory	1280 bytes (nickel delay lines)	32 kB (16 kB I + 16kB D L1 cache)
Secondary memory	1.2 MB (300,000 word magnetic film)	8 GB (typical micro SD flash card - not included)
Output bandwidth	25 characters/s	373 MB/s (1080p60 HDMI)
Weight	3-6 tons	9 g
Size	21 cabinets, each 2m x 77cm x 77cm	65mm x 30mm x 5.4mm





Application

- Logic circuits are an important part of almost all modern products
- Computers: The brain, body and limbs of computer systems
 - everything in it except peripherals
- Embedded Systems: The brains that control the systems
 - (ex. avionics, auto electronics, biomedical systems, etc.)
- Digital Signal Processing (DSP):
 - (ex. smart phone, HDTV, digital camera, music players, etc.)

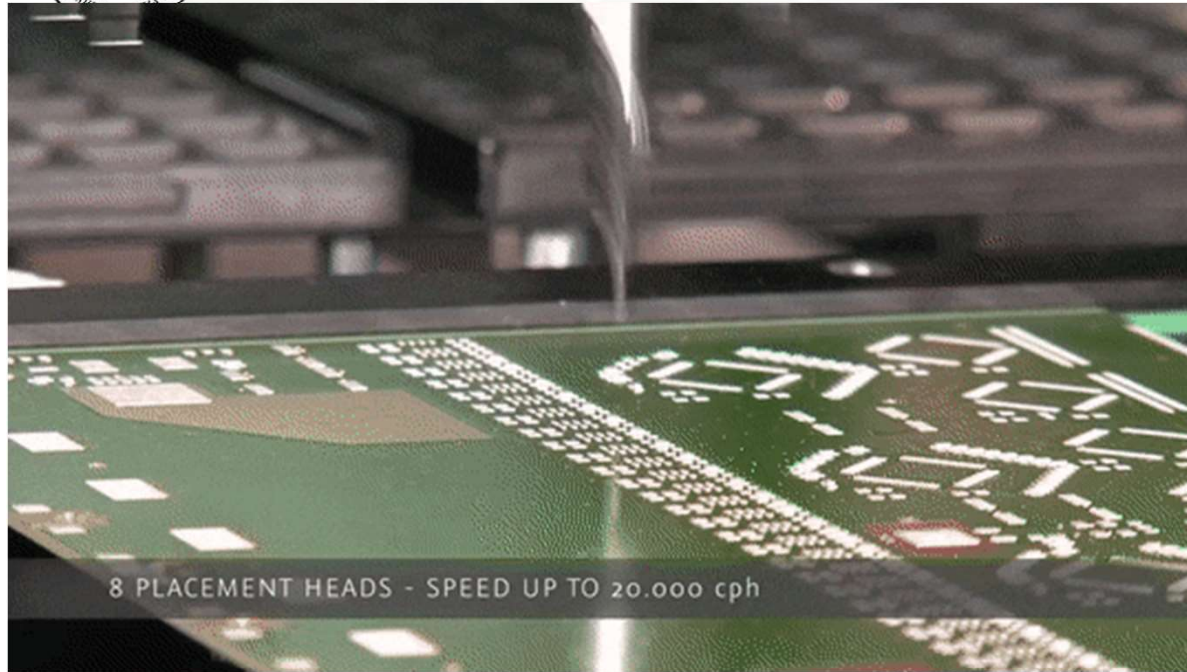


Top 30 innovations of the last 30 years to transform the world

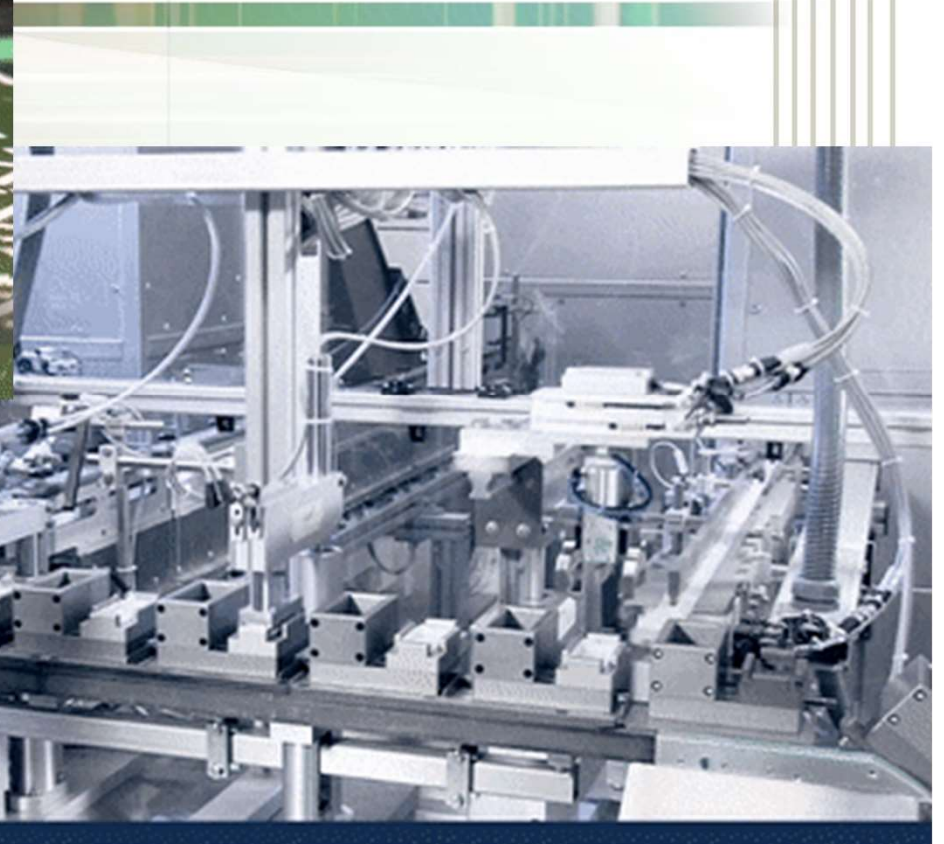
1. Internet, broadband, WWW (browser and html)
- 2. PC/laptop computers**
- 3. Mobile phones**
4. E-mail
5. DNA testing and sequencing/Human genome mapping
6. Magnetic Resonance Imaging (MRI)
- 7. Microprocessors**
8. Fiber optics
9. Office software (spreadsheets, word processors)
10. Non-invasive laser/robotic surgery (laparoscopy)
11. Open source software and services (e.g., Linux, Wikipedia)
- 12. Light emitting diodes**
- 13. Liquid crystal display (LCD)**
- 14. GPS systems**
15. Online shopping/ecommerce/auctions (e.g., eBay)
16. Media file compression (jpeg, mpeg, mp3)
17. Microfinance
- 18. Photovoltaic Solar Energy**
19. Large scale wind turbines
20. Social networking via the Internet
21. Graphic user interface (GUI)
- 22. Digital photography/videography**
- 23. RFID and applications (e.g., EZ Pass)**
24. Genetically modified plants
25. Bio fuels
26. Bar codes and scanners
27. ATMs
28. Stents
- 29. SRAM flash memory**
30. Anti retroviral treatment for AIDS



Programmable Logic Devices



8 PLACEMENT HEADS - SPEED UP TO 20.000 cph



Logic, SOC2020

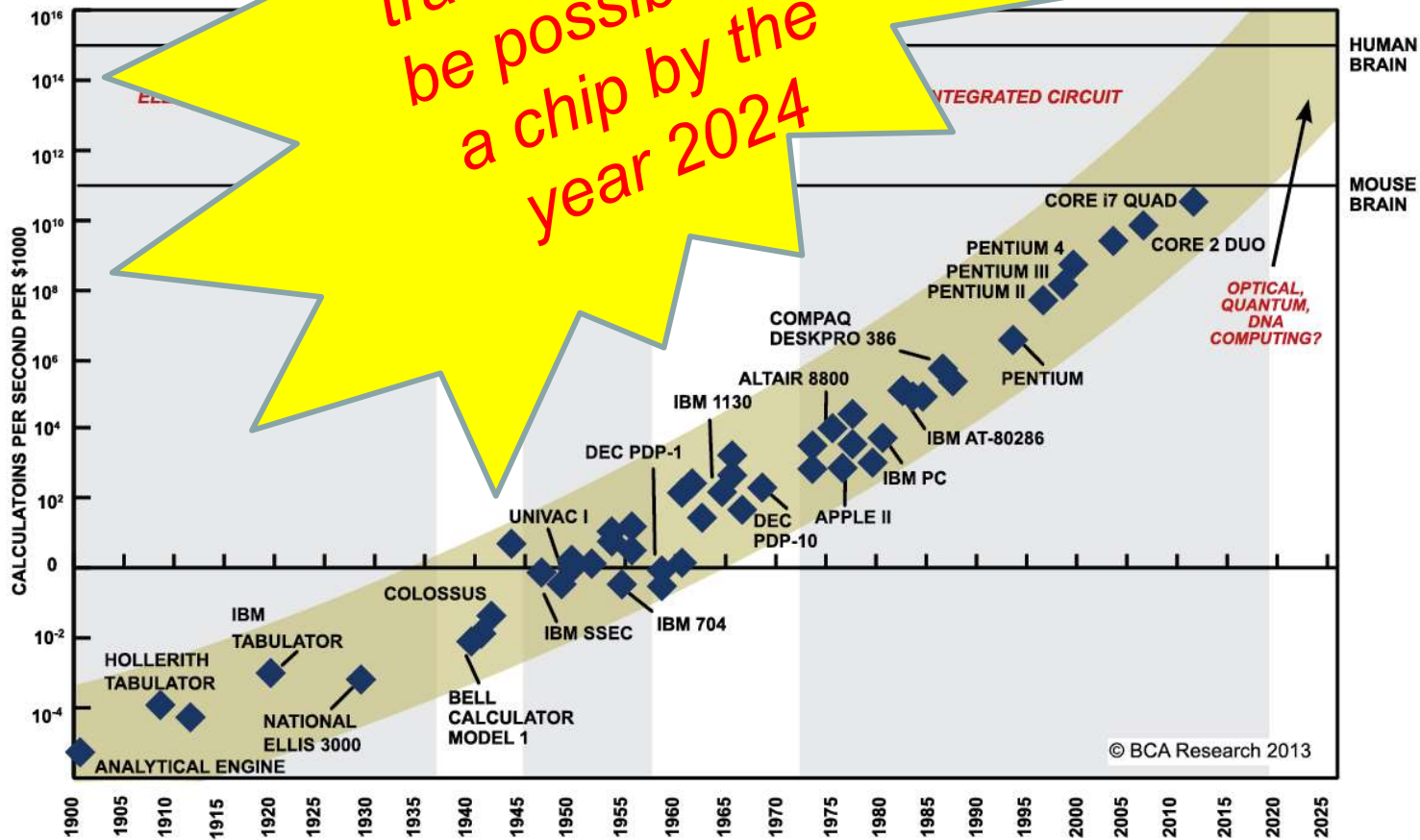


Digital Hardware

- By 1970 it was possible to place all the necessary circuitry to make a microprocessor on a chip.
 - Starting point of making PCs affordable
- Gordon Moore, chairman of Intel Corp., observed that number of transistors placed on a chip was doubling every 2 year
 - The phenomenon is known as *Moore's Law*

Digital Ha

100 billion transistors will be possible on a chip by the year 2024





Three main type of chips



Standard chips



Programmable logic devices



Custom chips



Standard chips

- Some commonly-used logic circuits
- Each standard chip contains a small amount of circuitry and performs a simple function
- Standard chips were popular for building logic circuits until the early 1980s
- Disadvantages:
 - Inefficient to use valuable space on PCBs for chips with low functionality
 - The functionality of each chip is fixed and cannot be changed.



Programmable logic devices

- These chips have a very general structure
- Include a collection of programmable switches –can be configured in many different ways
- Programmed by the end user, not by the manufacturer
 - Such chips are known as programmable logic devices (PLDs)
- Field-programmable gate array (FPGA) : most commonly-used type of PLD
 - The largest FPGAs contain billions of transistors, and support the implementation of complex digital systems.
- FPGAs are widely used commercially nowadays
- Disadvantages:
 - Consume valuable chip area
 - Limit the speed of operation of implemented circuits



Custom-Designed Chips

- In some cases, FPGAs may not meet the **desired performance** or **cost objectives**.
- A chip is designed first, and then the chip is manufactured by a company that has the fabrication facilities
 - Custom design or semi-custom design
- Application-specific integrated circuits : ASICs
- Main advantage : the design is optimized for a specific task
 - Usually better performance
- A disadvantage :
 - A considerable amount of time for manufacturing a custom chip

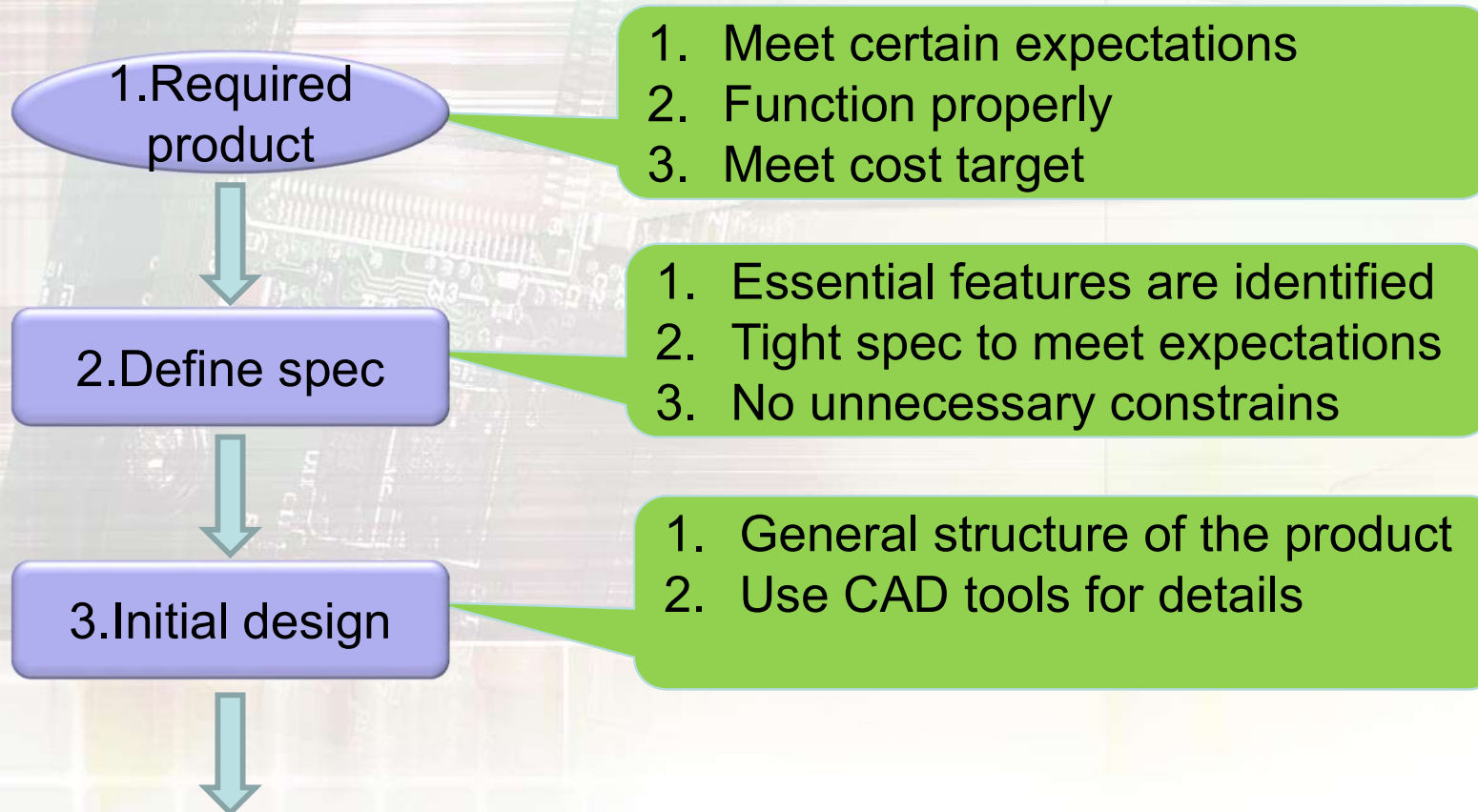
Examples for custom chips



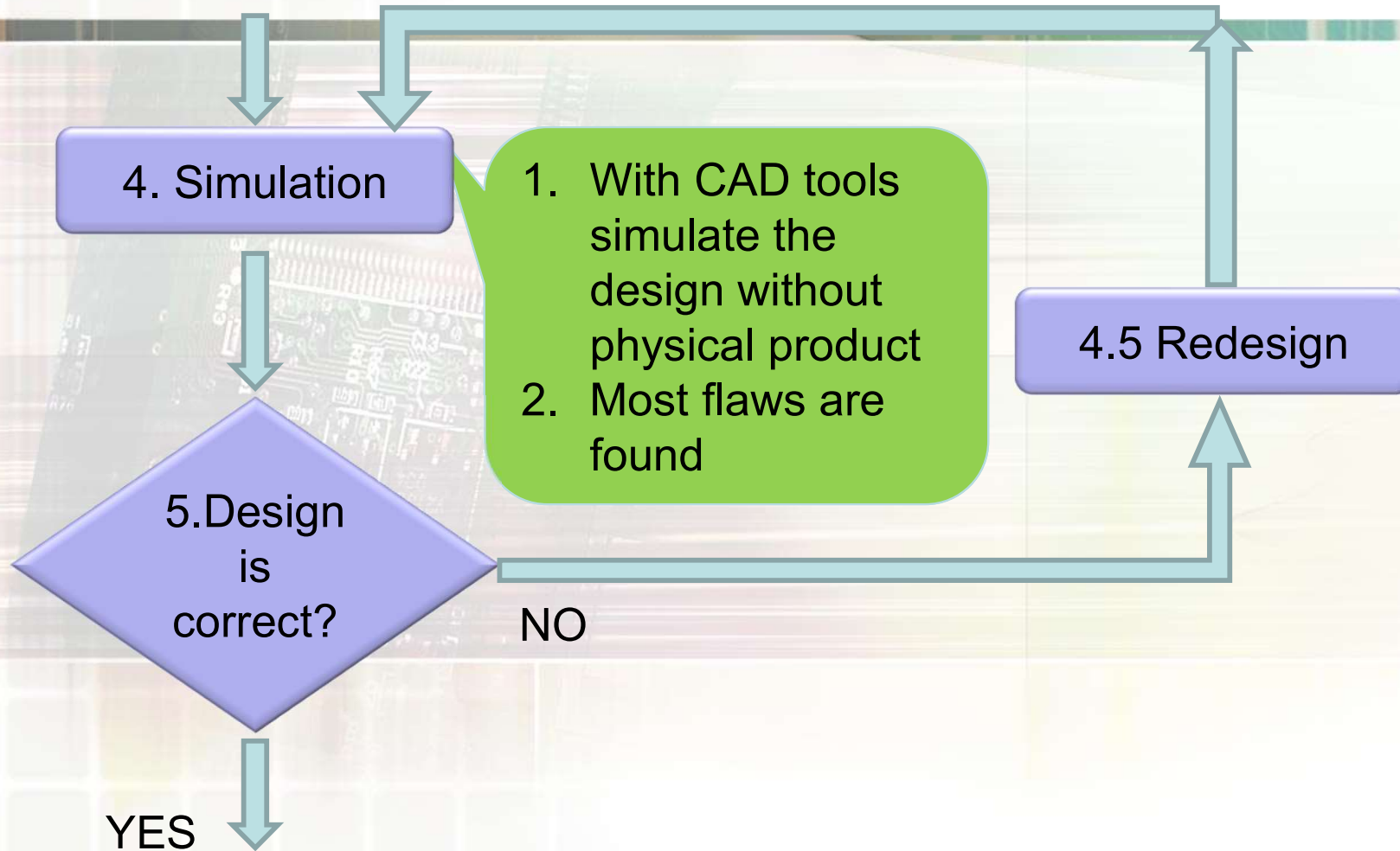
High Speed FIFO
Field Memory



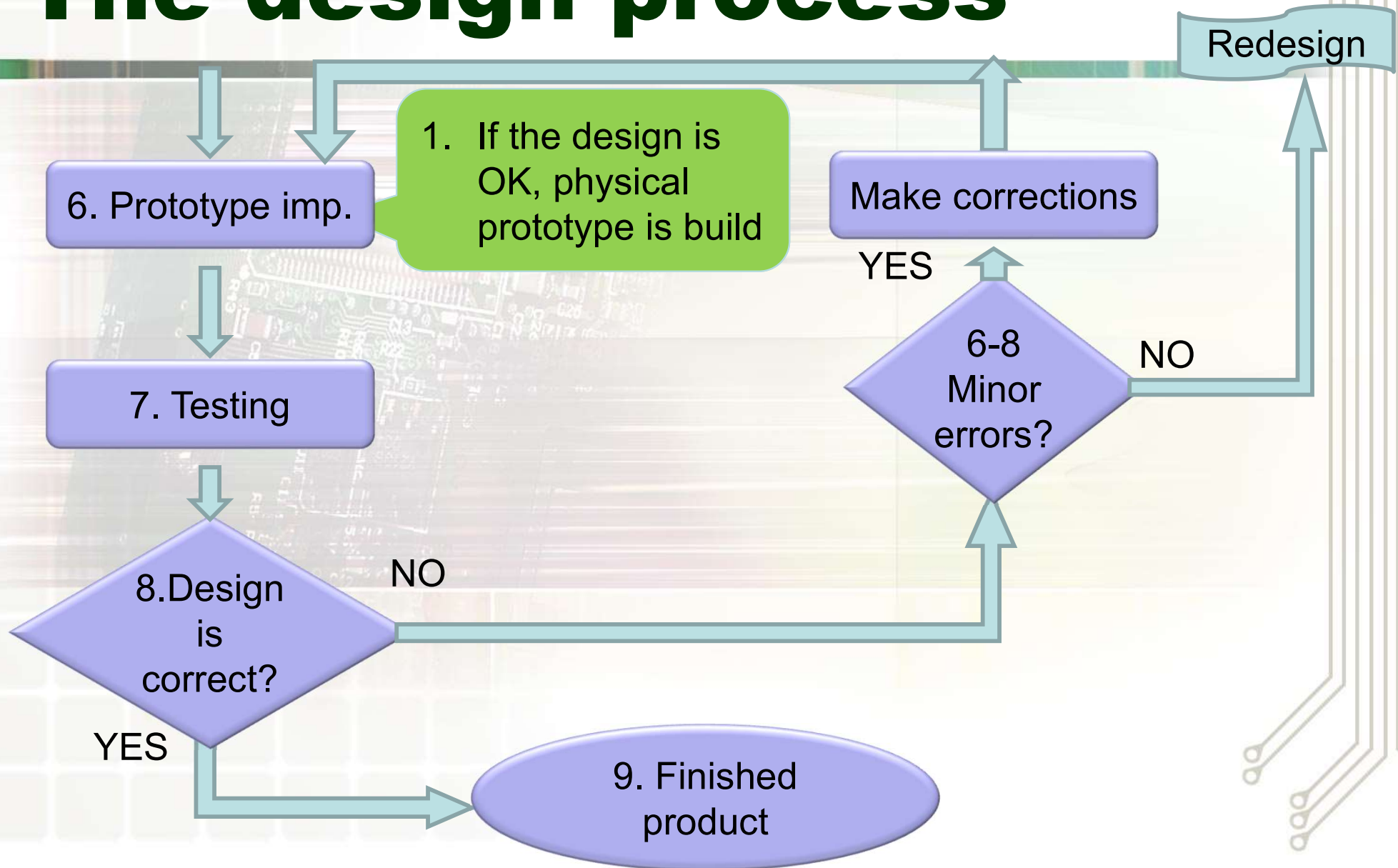
The design process



The design process



The design process



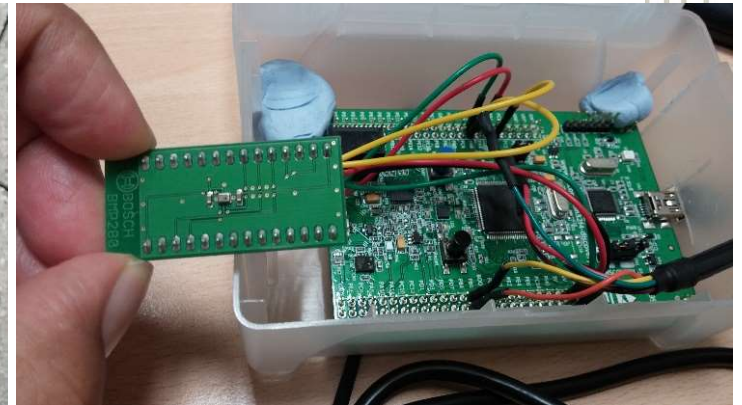
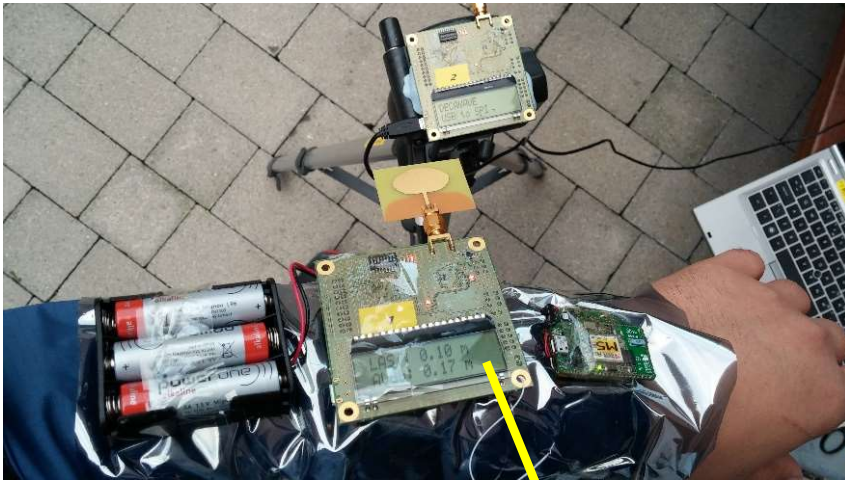


The design process-Practical

- Product: Sensor with wireless communication and onboard sensors
- Spec:
 - Powerful MCU
 - Wireless comm. Chips
 - Memory
 - Battery management
 - 9 Axis motion sensor
 - Barometer
 - Temperature sensor

The design process-Practical

- Initial design

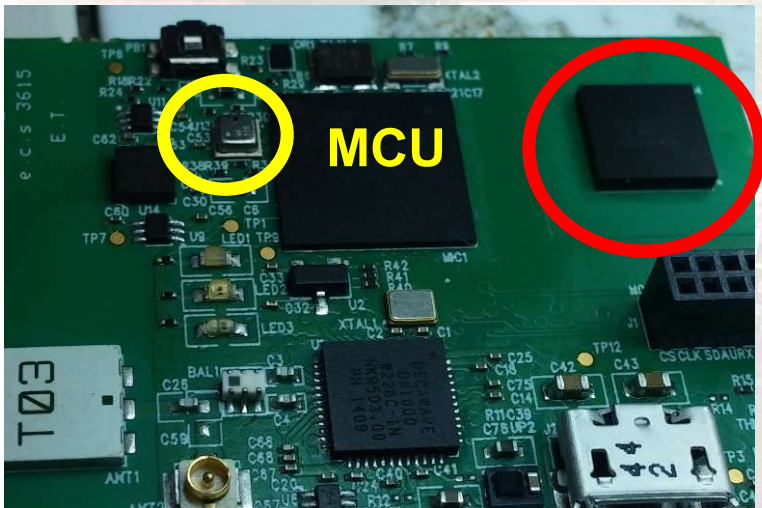


The design process-Practical

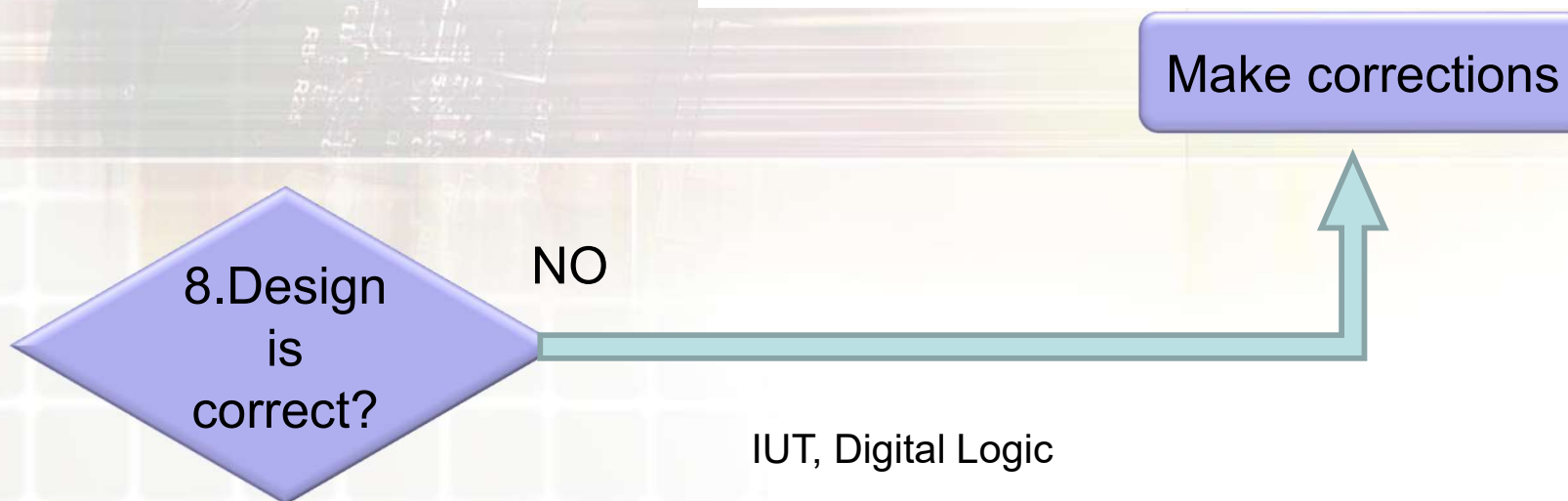
- Simulations are done for algorithms not for hardware!!!
- Prototype:
- Size: 3x4cm

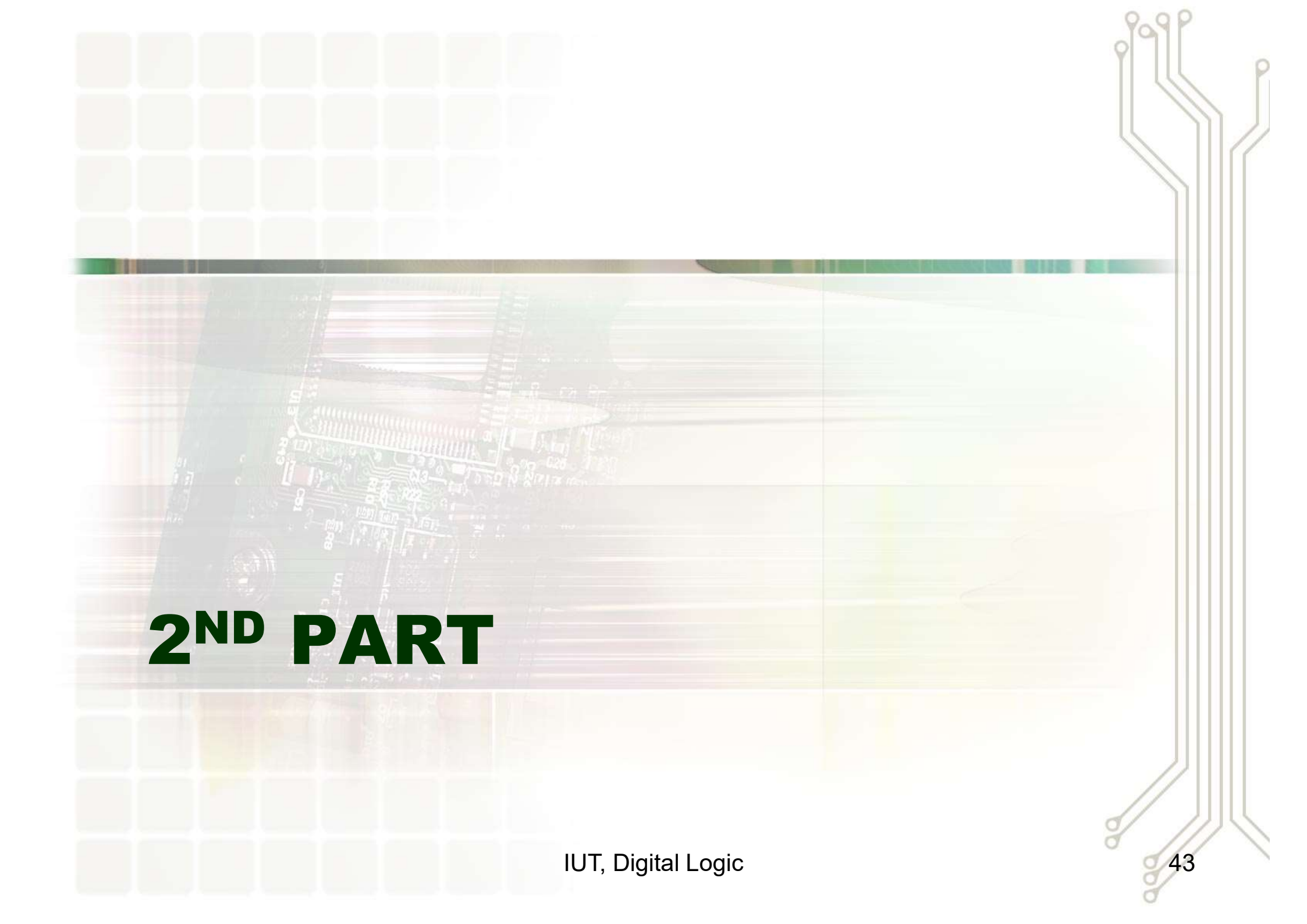


The design process-Practical



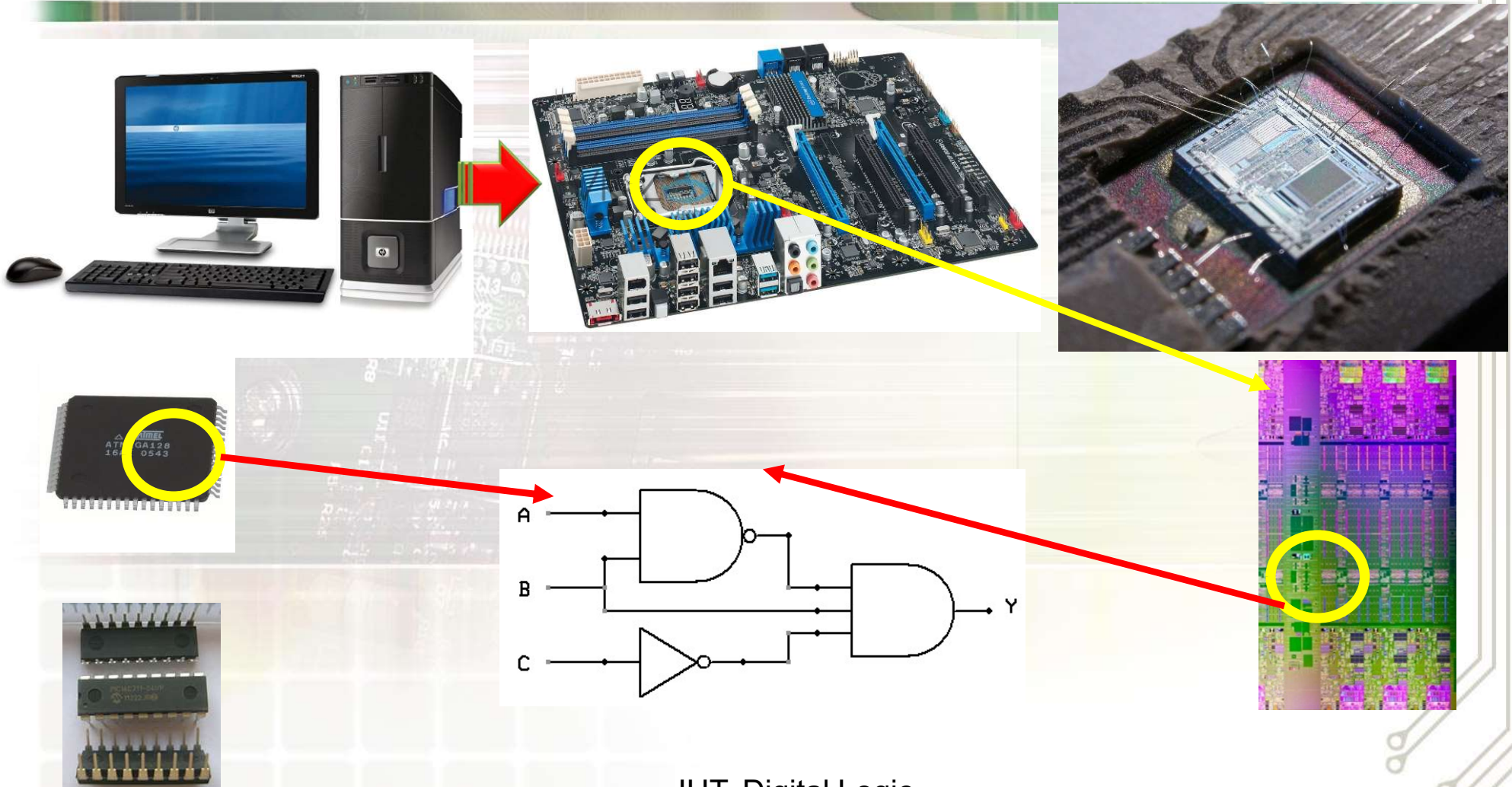
- Yellow – Temperature sensor
- Red – Bluetooth Chip





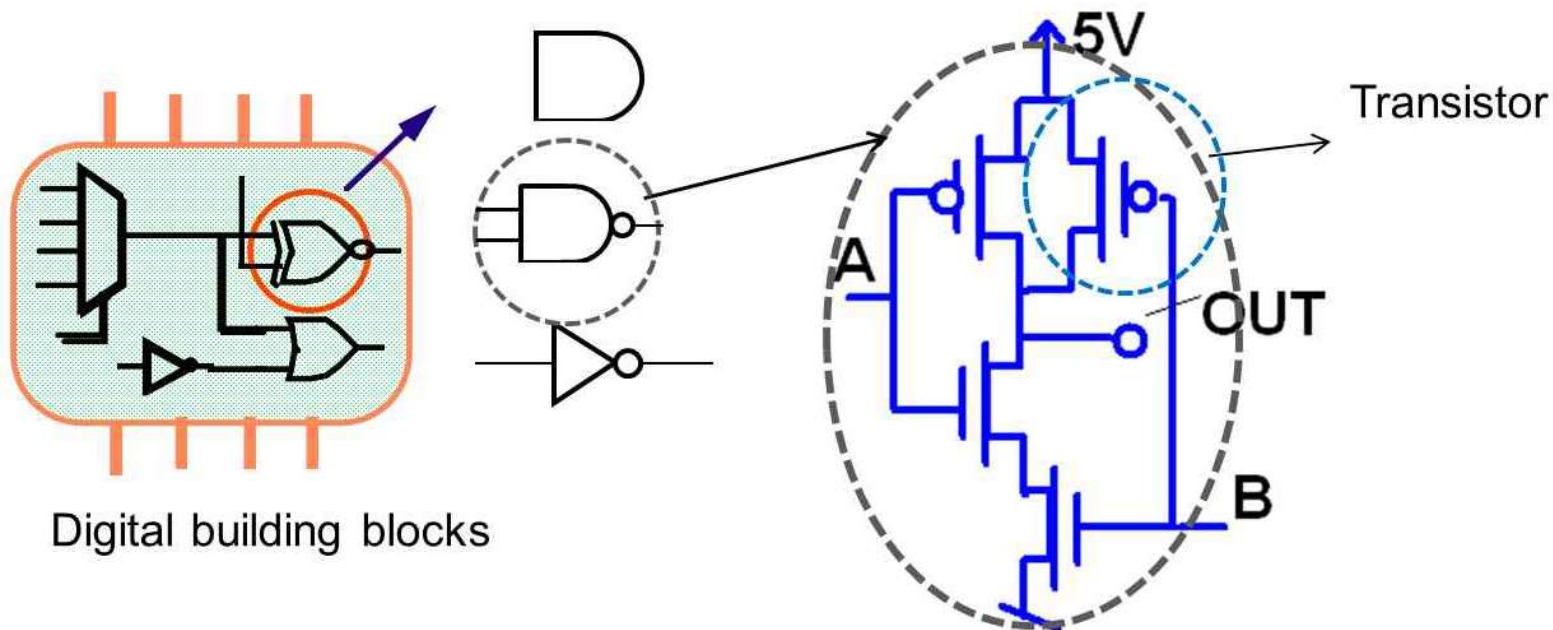
2ND PART

Structure of a Computer



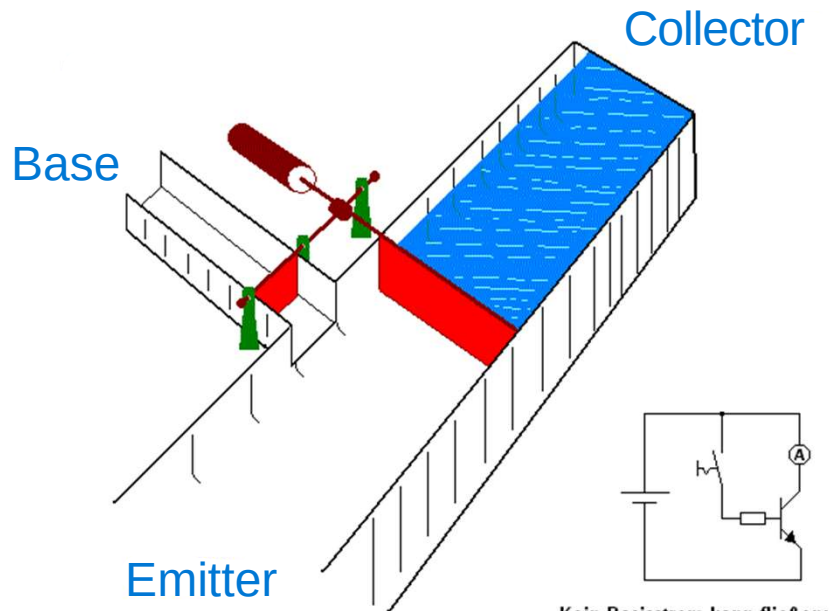
What are Logic Gates built from?

- Transistors ... the workhorse of every electronic device

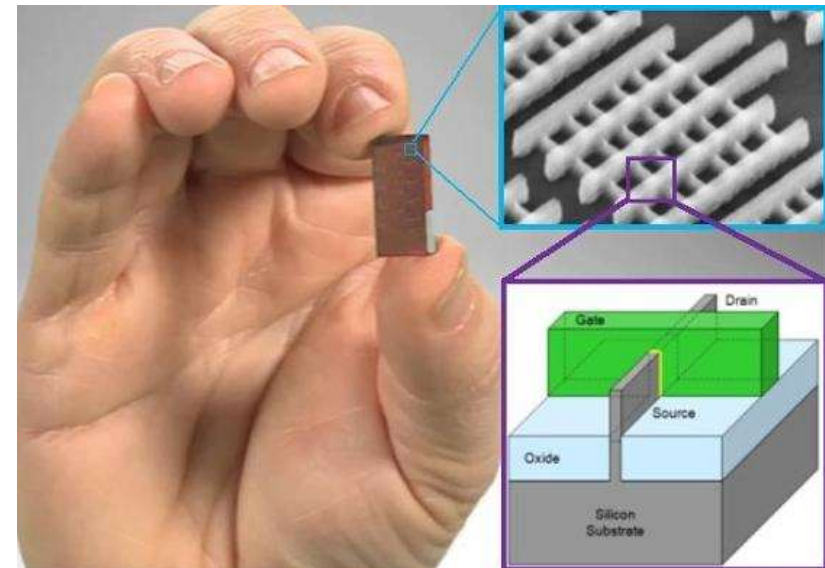


What is a Transistor?

- A **transistor** is a semiconductor device used to amplify or switch electronic signals and electrical power.



Kein Basisstrom kann fließen;
Der Transistor ist gesperrt.





Binary numbers

- Familiar decimal system numbers consists of digits with 10 possible values
 - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
 - Each digit represents a multiple of a power of 10
 - $1991 = 1 \times 10^3 + 9 \times 10^2 + 9 \times 10^1 + 1 \times 10^0$
- Positional number representation
 - $D = d_{n-1}d_{n-2} \dots d_1d_0$
 - $V(D) = d_{n-1} \times 10^{n-1} + d_{n-2} \times 10^{n-2} + \dots + d_1 \times 10^1 + d_0 \times 10^0$



Binary numbers

- Decimal numbers are *base 10* numbers
- For example, the number 8547 represents

$$8 \times 10^3 + 5 \times 10^2 + 4 \times 10^1 + 7 \times 10^0$$

- Since digital circuits represent information using only the values 0 and 1, it is not practical to have other digits
 - So we use binary, or *base-2*, system
 - $B = b_{n-1}b_{n-2} \dots b_1b_0$
 - $V(B) = b_{n-1} \times 2^{n-1} + b_{n-2} \times 2^{n-2} + \dots + b_1 \times 2^1 + b_0 \times 2^0$



Binary numbers

- Example:
 - What value does the binary number 1101 represent?
 - $V = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13$
- Base is indicated as a subscript
 - To specify binary number we write $(1101)_2$
- The range of integers that can be represented by binary system depends on number of bits
 - Using n bits we can represent numbers from 0 to $2^n - 1$



Binary numbers

	DECIMAL	BINARY	
10^0	0	0	2^0
	1	1	
	2	10	2^1
	3	11	
	4	100	2^2
	5	101	
	6	110	
	7	111	
	8	1000	2^3
	9	1001	
10^1	10	1010	
	11	1011	
	12	1100	
	13	1101	
	14	1110	
	15	1111	
	16	10000	2^4
	17	10001	
	18	10010	
	19	10011	
	20	10100	

1 1 1 0 1 0 0 0 1 1 1 0 1 0 0

MSB

LSB

- Group of bits together called *nibble*
- Group of 8 bits are called - *byte*



Conversion between Decimal and Binary systems

- A binary number is converted into decimal by binary number representation value equation
 - Evaluate it with decimal arithmetic
- Decimal number is converted into binary by constructing the number by using power of 2
 - $(17)_{10} = 2^4 + 2^0 = (10001)_2$
 - $(50)_{10} = 2^5 + 2^4 + 2^1 = (110010)_2$
- The conversion can be done by successively dividing the decimal number by 2



Conversion between Decimal and Binary systems

	Remainder	
• $765 / 2 = 382$	1	LSB
• $428 / 2 = 191$	0	
• $191 / 2 = 95$	1	
• $95 / 2 = 47$	1	
• $47 / 2 = 23$	1	
• $23 / 2 = 11$	1	
• $11 / 2 = 5$	1	
• $5 / 2 = 2$	1	
• $2 / 2 = 1$	0	MSB
• Result is $(1011111101)_2$		

Octal/Hex number systems

📄 The octal number system [from Greek: ΟΚΤΩ].

– Its base is 8 → eight digits 0, 1, 2, 3, 4, 5, 6, 7

✓ $(236.4)_8 = (?)_{10}$

$$2 \times 8^2 + 3 \times 8^1 + 6 \times 8^0 + 4 \times 8^{-1} = 158.5$$

📄 The hexadecimal number system [from Greek: ΔΕΚΑΕΞΙ].

– Its base is 16 → first 10 digits are borrowed from the decimal system and the letters A, B, C, D, E, F are used for the digits 10, 11, 12, 13, 14, 15

✓ $(D63FA)_{16} = (?)_{10}$

$$13 \times 16^4 + 6 \times 16^3 + 3 \times 16^2 + 15 \times 16^1 + 10 \times 16^0 = 877562$$



Conversion between Decimal and Binary systems

- Class exercise:

- a) $(1111010111)_2 = (983)_{10}$

- e)

- b) $(1000\ 0001)_2$

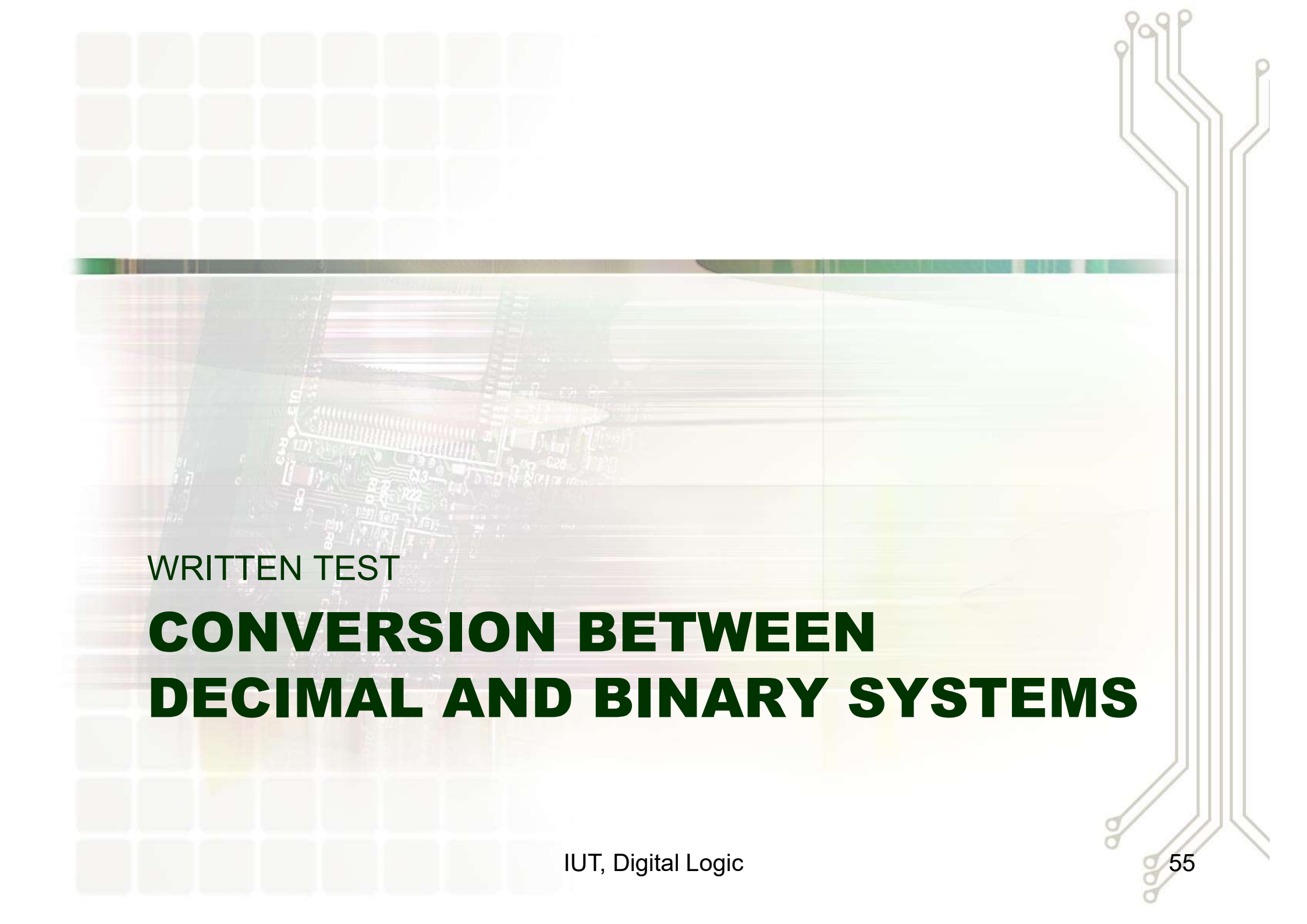
$(15026)_{10}$ $(0011101010110010)_2$

- c) $(42)_{10}$

- f)

- d) $(0100\ 0000)_2$

$(427)_{10}$ $(0001\ 1010\ 1011)_2$



WRITTEN TEST

CONVERSION BETWEEN DECIMAL AND BINARY SYSTEMS



Conversion between Decimal and Binary systems

- Class exercise:

- a) $(416)_{10} = ()_8 ?$

- e) $(FAB)_{16} = ()_{10} ; ()_2 ?$

- b) $(321)_8 = ()_{10} ?$

- f) $(1001\ 1110\ 0010\ 1100)_2 =$
 $= ()_{16} ; ()_{10} ; ()_8 ?$

- c) $(70)_8 = ()_{10} ?$

- d) $(99)_{10} = ()_2 ?$



Homework

- Ch1:
 - 1.1, 1.2, 1.3, 1.6, 1.7, 1.8



The end

